

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – Scenario Based Questions

|                  |                                                                                   |               |                              |
|------------------|-----------------------------------------------------------------------------------|---------------|------------------------------|
| Course Code:     | DSA0404                                                                           | Course Title: | FUNDAMENTALS OF DATA SCIENCE |
| CO Assessed:     | CO4                                                                               | Assessment:   | Assessment Tool 1            |
| Weightage:       | 40% of CIA                                                                        | Total Marks:  | 100                          |
| CO3 Statement:   | To apply Statistical inferences such as testing of Hypothesis and Point estimates |               |                              |
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| Section / Batch: | 2024                                                                              | Date:         | 17-08-26                     |

### Code of Conduct

I M Poojitha Reddy - 192472352 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M Poojitha Reddy

| Criteria                | Weight age | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  | Marks |
|-------------------------|------------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|-------|
| Understanding of Case   | 25         | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |       |
| Application of Concepts | 25         | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |       |
| Analysis                | 20         | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |       |
| Solution Design         | 20         | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |       |
| Clarity                 | 10         | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |       |

# 1. Hypothesis Testing Using p-value

## Question:

A software company claims that its new application has an average response time of less than 2 seconds. A hypothesis test gives a p-value of 0.004. At  $\alpha = 0.01$ , determine whether there is sufficient evidence to support the company's claim.

## Answer:

### Given:

- Claimed average response time = 2 seconds
- Company's claim: Average response time is **less than 2 seconds**
- p-value = 0.004
- Significance level,  $\alpha = 0.01$
- 

### Step 1: State the Hypotheses

Let  $\mu$  represent the true average response time of the application.

#### Null Hypothesis ( $H_0$ ):

$$\mu \geq 2 \text{ seconds}$$

#### Alternative Hypothesis ( $H_1$ ):

$$\mu < 2 \text{ seconds}$$

This is a **left-tailed test** because the company is claiming that the response time is less than 2 seconds.

### Step 2: Compare p-value with $\alpha$

The decision rule is:

- If p-value  $\leq \alpha \rightarrow$  Reject  $H_0$

- If  $p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$

Here,

$$p\text{-value} = 0.004$$

$$\alpha = 0.01$$

Since,

$$0.004 < 0.01$$

**we reject the null hypothesis.**

### Step 3: Interpretation

The p-value of 0.004 indicates that, assuming the null hypothesis is true, there is only a **0.4% chance** of obtaining a result as extreme as the observed result.

Since this probability is smaller than the chosen significance level of 1%, the observed evidence is statistically significant.

| Item                            | Value                     |
|---------------------------------|---------------------------|
| Claimed mean response time      | 2 seconds                 |
| Company's claim                 | $\mu < 2 \text{ seconds}$ |
| p-value                         | 0.004                     |
| Significance level ( $\alpha$ ) | 0.01                      |

### Hypotheses:

- $H_0: \mu \geq 2 \text{ seconds}$
- $H_1: \mu < 2 \text{ seconds}$

### Decision Rule:

**Condition Decision**

$p\text{-value} \leq \alpha$  Reject  $H_0$

$p\text{-value} > \alpha$  Fail to reject  $H_0$

**Conclusion:**

There is sufficient statistical evidence at the 1% significance level to support the company's claim that the new application has an average response time of less than 2 seconds.

Therefore, the company's claim is supported by the hypothesis test.

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## 2. Hypothesis Testing Using p-value

**Question:**

A digital marketing team launches a new advertisement and wants to know whether it increases the click-through rate. Before the campaign, the rate was 6%. After the campaign, a hypothesis test produces a p-value of 0.12. What should the data science team conclude at the 5% significance level?

**Answer:****Given:**

- Original click-through rate = 6%
- The team wants to determine whether the new advertisement **increases** the click-through rate.
- $p\text{-value} = 0.12$
- Significance level,  $\alpha = 0.05$

Let  $\mu$  represent the true click-through rate after the advertisement.

**Step 1: State the Hypotheses****Null Hypothesis ( $H_0$ ):**

The advertisement does not increase the click-through rate.

$$H_0: p \leq 0.06$$

### **Alternative Hypothesis ( $H_1$ ):**

The advertisement increases the click-through rate.

$$H_1: p > 0.06$$

This is a **right-tailed test** because we are testing for an increase.

### **Step 2: Compare p-value with $\alpha$**

Decision rule:

- If  $p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$
- If  $p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$

Given:

$$p\text{-value} = 0.12$$

$$\alpha = 0.05$$

Therefore,

$$0.12 > 0.05$$

Hence, we **fail to reject the null hypothesis**.

### **Step 3: Interpretation**

The p-value of 0.12 means that, assuming there is actually no increase in the click-through rate, there is a **12% probability** of obtaining a result at least as extreme as the observed result.

A 12% probability is greater than the allowed significance level of 5%. Therefore, the evidence is not strong enough to establish that the advertisement has increased the click-through rate.

This does **not** mean that the advertisement definitely has no effect. It only means that the available data does not provide sufficient statistical evidence to prove an increase.

### Conclusion:

- At the **5% significance level**, there is **insufficient statistical evidence** to conclude that the new advertisement increased the click-through rate from 6%.
  - Therefore, the data science team should **not claim a statistically significant improvement** based on this hypothesis test.
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## 3. Hypothesis Testing Using p-value

### Question:

A machine-learning team develops a new classification model. The old model has an accuracy of 82%, while the new model achieves 85%. A statistical test gives a p-value of 0.01. Explain what the p-value means and whether the team should conclude that the new model performs significantly better.

### Answer:

### Given:

- Accuracy of old model = 82%
- Accuracy of new model = 85%
- p-value = 0.01
- Significance level = 5% = 0.05

The team wants to determine whether the new model performs **significantly better** than the old model.

### Step 1: State the Hypotheses

Let  $\mu_1$  represent the performance of the old model and  $\mu_2$  represent the performance of the new model.

### Null Hypothesis ( $H_0$ ):

The new model does not perform significantly better than the old model.

$$H_0: \mu_2 \leq \mu_1$$

**Alternative Hypothesis ( $H_1$ ):**

The new model performs significantly better than the old model.

$$H_1: \mu_2 > \mu_1$$

This is a **right-tailed test**.

## **Step 2: Understand the p-value**

The p-value is **0.01**.

A p-value represents the probability of obtaining a result as extreme as, or more extreme than, the observed result **assuming that the null hypothesis is true**.

Therefore, a p-value of 0.01 means that if there were actually no significant improvement in the new model, there would be approximately a **1% chance** of observing a difference this large or larger due to random variation.

A small p-value provides strong evidence against the null hypothesis.

## **Step 3: Compare p-value with $\alpha$**

At the 5% significance level:

$$\alpha = 0.05$$

$$\text{p-value} = 0.01$$

Since,

$$\mathbf{0.01 < 0.05}$$

**we reject  $H_0$ .**

## Step 4: Interpretation

The new model has an observed accuracy of 85%, compared with 82% for the old model. The difference is 3 percentage points.

However, simply observing a higher accuracy is not enough to conclude that the improvement is statistically meaningful. The p-value helps determine whether this difference could reasonably be due to random chance.

Since the p-value is only 0.01, the evidence against the null hypothesis is statistically significant at the 5% level.

## Conclusion:

The team should **conclude that the new classification model performs significantly better** than the old model at the 5% significance level. Thus, the improvement from **82% to 85% is statistically significant**, based on the hypothesis test.

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# 4. Confidence Interval + p-value

## Question:

A factory claims that its defect rate is 2%. After randomly inspecting products, a 95% confidence interval for the defect rate is found to be (2.4%, 3.6%). At the same time, a hypothesis test gives a p-value of 0.01. Interpret both results and explain whether the company's claim should be rejected.

## Answer:

## Given:

- Company's claimed defect rate = 2%
- 95% Confidence Interval = (2.4%, 3.6%)
- p-value = 0.01
- Significance level = 5% = 0.05

We need to determine whether the factory's claim that the defect rate is 2% is supported by the data.



## Step 1: Interpretation of Confidence Interval

The 95% confidence interval for the true defect rate is:

**(2.4%, 3.6%)**

This means that the data provides a plausible range of **2.4% to 3.6%** for the true defect rate.

The company claims that the defect rate is **2%**.

However, 2% is **not contained within the confidence interval**.

The entire confidence interval lies above 2%.

Therefore, the sample data suggests that the actual defect rate is likely **higher than the company's claimed 2%**.

## Step 2: Hypothesis Testing

The hypotheses can be stated as:

**Null Hypothesis ( $H_0$ ):**

$$p = 0.02$$

**Alternative Hypothesis ( $H_1$ ):**

$$p \neq 0.02$$

Given:

$$\text{p-value} = 0.01$$

$$\alpha = 0.05$$

Decision rule:

- If  $\text{p-value} \leq \alpha \rightarrow \text{Reject } H_0$
- If  $\text{p-value} > \alpha \rightarrow \text{Fail to reject } H_0$

Since,

$$0.01 < 0.05$$

we reject  $H_0$ .

| Method   | Result       | Interpretation         |
|----------|--------------|------------------------|
| 95% CI   | (2.4%, 3.6%) | Does not contain 2%    |
| p-value  | 0.01         | Less than 0.05         |
| Decision | Reject $H_0$ | Claim is not supported |

### Step 3: Combining Both Results

Both statistical methods lead to the same conclusion.

The confidence interval is:

**2.4% to 3.6%**

and does not include the claimed value of 2%.

Similarly, the p-value is 0.01, which is less than the 5% significance level.

Therefore, there is strong statistical evidence that the actual defect rate is different from, and in this case higher than, 2%.

### Conclusion:

The factory's claim that its defect rate is **2% should be rejected** at the 5% significance level. The statistical evidence suggests that the true defect rate is likely to be **higher than 2%**, with the 95% confidence interval estimating it to be between **2.4% and 3.6%**.

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## 5. Confidence Interval + p-value

### Question:

An online shopping company wants to determine whether a new user-interface design increases the average amount spent per customer. The data scientist calculates a 95% confidence interval for the difference in average spending as (₹120, ₹480) and obtains a p-value of 0.002. Interpret the confidence interval and p-value, and provide a business recommendation.

### Answer:

### Given:

- 95% Confidence Interval for difference in average spending = (₹120, ₹480)
- p-value = 0.002
- Significance level = 5% = 0.05

The objective is to determine whether the new user-interface design increases the average spending per customer.

### Step 1: Interpretation of Confidence Interval

The 95% confidence interval for the difference in average spending is:

**(₹120, ₹480)**

This indicates that the estimated increase in average customer spending is between **₹120 and ₹480 per customer**.

Most importantly, the entire confidence interval is **positive**.

The interval does not contain ₹0.

A difference of ₹0 would indicate that there is no change in average spending between the old and new interfaces.

Since ₹0 is not included in the interval, the data provides evidence that the new interface increases average customer spending.

### Step 2: Interpretation of p-value

The hypothesis test can be represented as:

### Null Hypothesis ( $H_0$ ):

The new UI does not increase average customer spending.

### Alternative Hypothesis ( $H_1$ ):

The new UI increases average customer spending.

Given:

$$p\text{-value} = 0.002$$

$$\alpha = 0.05$$

Since,

$$0.002 < 0.05$$

**we reject the null hypothesis.**

The p-value of 0.002 indicates that, if there were actually no increase in average spending, there would be only about a **0.2% probability** of observing evidence this strong or stronger due to random chance. Thus, there is strong statistical evidence of an increase.

### Step 3: Combined Interpretation

Both the confidence interval and p-value provide consistent evidence.

- The 95% confidence interval is **₹120 to ₹480**, entirely above ₹0.
- The p-value is **0.002**, which is much smaller than 0.05.
- Therefore, the increase is **statistically significant**.
- The estimated increase in spending is practically meaningful because even the lower bound of the interval is ₹120.

| Method   | Result       | Interpretation       |
|----------|--------------|----------------------|
| 95% CI   | ₹120 to ₹480 | Entirely positive    |
| p-value  | 0.002        | < 0.05               |
| Decision | Reject $H_0$ | Significant increase |

| Method          | Result                | Interpretation           |
|-----------------|-----------------------|--------------------------|
| Business action | Adopt/consider new UI | Higher spending expected |

### Business Recommendation:

- ✓ The company should consider implementing the new user-interface design because the statistical evidence strongly suggests that it increases average spending per customer.
- ✓ The company can expect the average spending to increase by approximately ₹120 to ₹480 per customer, based on the 95% confidence interval.
- ✓ However, before a full-scale implementation, the company should also consider practical factors such as development cost, customer experience, conversion rate, and long-term customer retention.

### Final Conclusion:

There is **strong statistical evidence that the new UI increases average customer spending**. Since the confidence interval is entirely positive and the p-value is significantly below 0.05, the company should favor adopting the new UI, provided that the implementation costs and other business factors are acceptable.